

A bit of theory as far as relevant to the implementation

The necessary theory - for non-reduced axes - is found in [Ryssens et al, PHYSICAL REVIEW C 92, 064318 \(2015\)](#) section III.B Lagrange-mesh representation.

Non-reduced axes

(As from [Ryssens et al, PHYSICAL REVIEW C 92, 064318 \(2015\)](#) section III.B Lagrange-mesh representation)

Grid points

Using M evenly space grid point at distance dx , the boundaries of the box are $[-\frac{M}{2}dx, \frac{M}{2}dx]$. The width of the box is $L = Mdx$. We require $M = 2N$ to be even such that there are $N = \frac{M}{2}$ grid points on each side of the origin and the origin is avoided as a grid point.

For non-reduced axes (where the shift parameter σ is required be zero) the grid points are:
[eq 1]

$$\pm\frac{1}{2}dx, \pm\frac{3}{2}dx, \dots, \pm\frac{N-3}{2}dx, \pm\frac{N-1}{2}dx$$

or

[eq 2]

$$x_{\pm i} = \pm(\frac{1}{2} + i)dx$$
$$i = 0..N-1$$

in order:

[eq 3]

$$-\frac{2N-1}{2}dx, -\frac{2N-3}{2}dx, \dots, -\frac{3}{2}dx, -\frac{1}{2}dx, \frac{1}{2}dx, \frac{3}{2}dx, \dots, \frac{2N-3}{2}dx, \frac{2N-1}{2}dx$$

or

[eq 4]

$$x_i = -\frac{2N-1-2i}{2} = \frac{1}{2} + i - N, i = 0..2N-1$$

Basis functions

The basis functions are plane waves:

[eq 5]

$$\phi_k(x) = \frac{1}{\sqrt{L}} \exp\left(\frac{2\pi}{L} kx\right)$$

$$k = \pm \frac{1}{2}, \pm \frac{3}{2}, \dots, \pm \frac{N-1}{2},$$

Interpolation

The Lagrange interpolation functions are:

[eq 6]

$$f_i(x) = \frac{1}{2N} \frac{\sin\left(\frac{\pi}{dx}(x - x_i dx)\right)}{\sin\left(\frac{\pi}{dx} \frac{x - x_i dx}{2N}\right)}$$

By construction, the Lagrange interpolation functions $f_i(x)$ have the property of being equal to 1 at the i -th mesh point, x_i , and 0 at all others:

[eq 7]

$$f_i(x_j) = \delta_{ij}$$

Note that the arguments of the two sine functions are the same, apart from a factor $1/2N$:

[eq 8]

$$f_i(x) = \frac{1}{2N} \frac{\sin(A_i(x))}{\sin\left(\frac{A_i(x)}{2N}\right)}, \quad A_i(x) = \frac{\pi}{dx}(x - x_i dx)$$

This can be exploited for efficiency in a computation.

An arbitrary function $h(x)$ taking the values $h_i = h(x_i)$ on the grid points can be interpolated on an arbitrary point $x \in [-Ndx, Ndx]$ as:

[ea 9]

$$h(x) = \sum_{i=0}^{2N-1} f_i h(x_i)(x)$$

This is essentially a dot product $\mathbf{f} \cdot \mathbf{h}$.

Derivatives

The 1st and 2nd order derivatives of the Lagrange interpolation functions are:

[eq 10]

$$D_{ji}^{(1)} = \left. \frac{df_i(x)}{dx} \right|_{x=x_j} = \begin{cases} (-1)^{i-j} \frac{\pi}{2Ndx} \frac{1}{\sin(\pi(i-j)/2N)}, & \text{for } i \neq j. \\ 0, & \text{for } i = j. \end{cases}$$

$$D_{ji}^{(2)} = \left. \frac{d^2 f_i(x)}{dx^2} \right|_{x=x_j} = \begin{cases} (-1)^{i-j+1} 2 \left(\frac{\pi}{2Ndx} \right)^2 \frac{\cos(\pi(i-j)/2N)}{\sin^2(\pi(i-j)/2N)}, & \text{for } i \neq j. \\ -\frac{\pi^2}{3dx^2} \left(1 - \frac{1}{(2N)^2} \right), & \text{for } i = j. \end{cases}$$

By applying these to the expansion $\phi(x) = \sum_{i=0}^{2N-1} \phi(x_i) f_i(x)$ we obtain

[eq 11]

$$\left. \frac{d\mathbf{h}}{dx} = \frac{dh(x)}{dx} \right|_{x=x_j} = \sum_{i=0}^{2N-1} \left. \frac{df_i(x)}{dx} \right|_{x=x_j} h(x_i) = \sum_{i=0}^{2N-1} D_{ji}^{(1)} h(x_i) = \mathbf{D}^{(1)} \mathbf{h}$$

So, the column vector $\frac{d\mathbf{h}}{dx}$ of the derivatives of $\mathbf{h}$ at all grid points is found as a matrix product of $\mathbf{D}^{(1)}$ with the column vector of the values of $\mathbf{h}$ at all the gridpoints. As the 2nd derivative forms a matrix as well, we also have

[eq 12]

$$\frac{d^2\mathbf{h}}{dx^2} = \mathbf{D}^{(2)} \mathbf{h}$$

Reduced axes

Grid points

On reduced axes only the N grid points to the right of the origin are kept:

[eq 13]

$$\frac{1}{2}dx, \frac{3}{2}dx, \dots, \frac{2N-3}{2}dx, \frac{2N-1}{2}dx$$

or

[eq 14]

$$x_i = (i + \frac{1}{2})dx$$

$$i = 0..N-1$$

(In the case of reduced axis a shift $\sigma < dx$ may be subtracted from each grid point.)

Interpolation

An arbitrary function $h(x)$, $x \in [0, Ndx]$ taking the values $h_i = h(x_i)$ on the grid points on a reduced axis can be interpolated as :

[eq 15]

$$h(x) = \sum_{i=0}^{2N-1} [h_i f_i(x) \pm h_i f_{-i}(x)]$$

where $+$, resp. $-$, is selected if $h(x)$ is symmetric, resp. antisymmetric, and $f_{-i}(x)$ is the Lagrange interpolation function corresponding to the i -th grid point to the left of the origin:

[eq 16]

$$f_{-i}(x) = \frac{1}{2N} \frac{\sin(\frac{\pi}{dx}(x - x_{-i}dx))}{\sin(\frac{\pi}{dx} \frac{x - x_{-i}dx}{2N})} = \frac{1}{2N} \frac{\sin(\frac{\pi}{dx}(x + (\frac{1}{2} + i)dx))}{\sin(\frac{\pi}{dx} \frac{x + (\frac{1}{2} + i)dx}{2N})}$$

This time we have a sum or a difference of two dot products $\mathbf{h} \cdot \mathbf{f}_+ \pm \mathbf{h} \cdot \mathbf{f}_-$

Again the arguments of the two sine functions are the same, apart from a factor $1/2N$.

Derivatives

The formula for the derivative of a function h expanded on a reduced grid is found easily by extending the column vector $\mathbf{h}$ (of length N) on the reduced grid as
[eq 17]

$$\begin{bmatrix} \pm \mathbf{g} \\ - - \\ \mathbf{h} \end{bmatrix}$$

with $\mathbf{g}$ containing the same elements as $\mathbf{h}$ but in reverse order, if the function h is symmetric, and its negatives in reverse order, if it antisymmetric as if they were mirrored by a horizontal line between the two, and then forming the matrix product
[eq 18]

$$\mathbf{D} \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = \begin{bmatrix} \mathbf{D}^- \\ \mathbf{D}^+ \end{bmatrix} \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = \begin{bmatrix} \mathbf{D}^- \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} \\ - - - - \\ \mathbf{D}^+ \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} \end{bmatrix}$$

The upper half of $\mathbf{D}$, $\mathbf{D}^-$, produces the derivatives on the negative x -axis, and the lower half of $\mathbf{D}$, $\mathbf{D}^+$, produces the derivatives on the positive x -axis. The upper part and the lower part are obviously related by symmetry: if the function $h(x)$ is symmetric, then the derivative is antisymmetric and *vice versa*. Thus we are only interested in the lower part. Writing $\mathbf{D}^+$ as $[\mathbf{D}_{ll} | \mathbf{D}_{lr}]$ (ll stands for lower-left and lr for lower-right quadrant of $\mathbf{D}$) we find:
[eq 19]

$$\mathbf{D}^+ \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = [\mathbf{D}_{ll} | \mathbf{D}_{lr}] \begin{bmatrix} \pm \mathbf{g} \\ \mathbf{h} \end{bmatrix} = [\mathbf{E}_{ll} | \mathbf{D}_{lr}] \begin{bmatrix} \pm \mathbf{h} \\ \mathbf{h} \end{bmatrix}$$

where $\mathbf{E}_{ll}$ is obtained from $\mathbf{D}_{ll}$ by reversing the order of the rows. Reversing the order of the rows of $\mathbf{D}_{ll}$ forces us to also reverse the order of the elements of $\pm \mathbf{g}$, which then becomes, by definition $\pm \mathbf{h}$. Finally, we obtain:
[eq 20]

$$\pm \mathbf{E}_{ll} \mathbf{h} + \mathbf{D}_{lr} \mathbf{h} = [\mathbf{D}_{lr} \pm \mathbf{E}_{ll}] \mathbf{h}$$

or

[eq 21]

$$\frac{d\mathbf{h}}{dx} = [\mathbf{D}_{lr}^{(1)} \pm \mathbf{E}_{ll}^{(1)}] \mathbf{h}$$

where we have reintroduced the superscript ⁽¹⁾ to indicate the first order derivative. By the same reasoning, we obtain for the 2nd derivative:
[eq 22]

$$\frac{\mathbf{d}^2\mathbf{h}}{\mathbf{d}\mathbf{x}^2} = [\mathbf{D}_{lr}^{(2)} \pm \mathbf{E}_l^{(2)}]\mathbf{h}$$

***N*-dimensional grids**

The full Cartesian 3D representation of a function $h(\mathbf{r})$, where $\mathbf{r} = [x \ y]$ (2D case), or $\mathbf{r} = [x \ y \ z]$ (3D case), is then provided (for the 3D case) by
[eq 23]

$$h(\mathbf{r}) = \sum_{ijk} h_{ijk} f_i(x) f_j(y) f_k(z)$$

where the number of discretization points does not have to be the same in each direction. In this case, the derivative matrices $\mathbf{D}^{(1)}$ and $\mathbf{D}^{(2)}$ have to be set up separately for each direction, taking into account whether the axis is reduced or not.